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| Team name: | PRIMEX FCoEE |
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|                            |                           |                                |
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| Name of the Local Currency | Total Cost Local Currency | Total Cost (\$) U.S.A. Dollars |
| Malaysian Ringgit MYR      | RM3,232.00                | \$808.00                       |

| #     | Component                 | Part Name                                   | Author / Manufacturer | Source     | Qty | Unit Cost (MYR) | Unit Cost (USD) | Total Cost (MYR) | Total Cost (USD) |
|-------|---------------------------|---------------------------------------------|-----------------------|------------|-----|-----------------|-----------------|------------------|------------------|
| 1     | High-Level Processor      | Jetson Orin Nano 8GB RAM + 512GB SSD        | NVIDIA                | cytron.io  | 1   | RM2,000.00      | \$500.00        | RM2,000.00       | \$500.00         |
| 2     | Low-Level Processor       | ESP32-S3 DevKitC-1                          | Espressif             | Shopee     | 1   | RM50.00         | \$12.50         | RM50.00          | \$12.50          |
| 3     | Distance Sensor           | VL53L0X V2 Time-of-Flight Sensor            | STMicroelectronics    | cytron.io  | 4   | RM25.00         | \$6.25          | RM100.00         | \$25.00          |
| 4     | Camera Module             | IMX219 160° Wide Angle Camera for Jetson    | Arducam               | cytron.io  | 2   | RM85.00         | \$21.25         | RM170.00         | \$42.50          |
| 5     | Colour Sensor             | mBlock mBot2 Colour Sensor                  | Makeblock             | cytron.io  | 1   | RM65.00         | \$16.25         | RM65.00          | \$16.25          |
| 6     | DC Motor with Encoder     | JGB37-520 12V 76RPM Hall Encoder DC Motor   | JGB                   | Shopee     | 4   | RM35.00         | \$8.75          | RM140.00         | \$35.00          |
| 7     | Motor Driver              | L298N Dual H-Bridge Motor Driver            | ST / Generic          | Shopee     | 2   | RM6.00          | \$1.50          | RM12.00          | \$3.00           |
| 8     | Servo Motor               | MG996R Metal Gear Servo Motor               | TowerPro              | Shopee     | 1   | RM18.00         | \$4.50          | RM18.00          | \$4.50           |
| 9     | Main Battery              | 20V 1.5Ah Power Tool Battery (Keelat)       | Keelat                | Shopee     | 2   | RM50.00         | \$12.50         | RM100.00         | \$25.00          |
| 10    | Battery Charger           | 20V Universal Battery Charger (Keelat)      | Keelat                | Shopee     | 1   | RM20.00         | \$5.00          | RM20.00          | \$5.00           |
| 11    | Battery Dock              | Power Tool Battery Dock / Holder            | Generic               | Shopee     | 1   | RM15.00         | \$3.75          | RM15.00          | \$3.75           |
| 12    | 5V Regulator              | 5V USB Step-Down Regulator (Max 24V in)     | Generic               | Shopee     | 1   | RM10.00         | \$2.50          | RM10.00          | \$2.50           |
| 13    | 12V Regulator             | 12V 5A Step-Down Regulator (Max 24V in)     | Generic               | Shopee     | 1   | RM15.00         | \$3.75          | RM15.00          | \$3.75           |
| 14    | Start Button              | Momentary Push Button (Green, 19mm)         | Generic               | Shopee     | 1   | RM5.00          | \$1.25          | RM5.00           | \$1.25           |
| 15    | Stop Button               | Emergency Stop Push Button (Red, 19mm)      | Generic               | Shopee     | 1   | RM20.00         | \$5.00          | RM20.00          | \$5.00           |
| 16    | Power Button              | Self-Latch Push Button (Blue, 19mm)         | Generic               | Shopee     | 1   | RM10.00         | \$2.50          | RM10.00          | \$2.50           |
| 17    | RGB LED Indicator         | 10mm RGB LED Module (Common Anode)          | Generic               | Shopee     | 1   | RM6.00          | \$1.50          | RM6.00           | \$1.50           |
| 18    | Data Cable                | USB Type-C to Type-A Cable                  | Generic               | Local Shop | 2   | RM8.00          | \$2.00          | RM16.00          | \$4.00           |
| 19    | 3D Filament - Body        | ABS-CF Carbon Fibre Reinforced ABS 1kg      | Generic               | Local Shop | 1   | RM150.00        | \$37.50         | RM150.00         | \$37.50          |
| 20    | 3D Filament - Frame       | PETG-CF Carbon Fibre Reinforced PETG 1kg    | Generic               | Local Shop | 1   | RM120.00        | \$30.00         | RM120.00         | \$30.00          |
| 21    | 3D Filament - Accessories | PLA Filament 1kg                            | Generic               | Local Shop | 1   | RM60.00         | \$15.00         | RM60.00          | \$15.00          |
| 22    | Cables & Electronics      | Miscellaneous Cables, Connectors, Resistors | Various               | Local Shop | 1   | RM100.00        | \$25.00         | RM100.00         | \$25.00          |
| 23    | Fasteners                 | Screws, Nuts, Standoffs, Washers            | Various               | Local Shop | 1   | RM30.00         | \$7.50          | RM30.00          | \$7.50           |
| TOTAL |                           |                                             |                       |            |     |                 |                 | RM3,232.00       | \$808.00         |

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| Name of the Local Currency | Total Cost Local Currency | Total Cost (\$) U.S.A. Dollars |
|----------------------------|---------------------------|--------------------------------|
| Malaysian Ringgit MYR      | RM0.00                    | \$0.00                         |

| #  | Name                   | Type          | Description                                                                             | Source                                                                    | Author                                      | Cost (MYR)       | Cost (USD)       |
|----|------------------------|---------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------|------------------|------------------|
| 1  | OpenCV (cv2)           | Library       | Image pre-processing, contour detection, HoughCircles, HSV thresholding, camera capture | <a href="https://opencv.org">opencv</a>                                   | OpenCV.org                                  | FREE             | FREE             |
| 2  | TensorFlow / Keras     | Library       | CNN model training, saved model loading, live inference for Greek letter classification | <a href="https://tensorflow.org">tensorflow.org</a>                       | Google / TensorFlow Dev                     | FREE             | FREE             |
| 3  | NumPy                  | Library       | Array masking for per-ring colour sampling and numerical computation                    | <a href="https://numpy.org">numpy.org</a>                                 | NumPy Developers                            | FREE             | FREE             |
| 4  | PySerial               | Library       | USB serial communication with ESP32-S3 for commands and sensor status                   | <a href="https://pypi.org/project/pyserial">pypi.org/project/pyserial</a> | Chris Liechti                               | FREE             | FREE             |
| 5  | VL53L0X Python Library | Library       | I2C distance sensor reading with XSHUT multiplexing for all four ToF sensors            | <a href="https://github.com/pololu/vl53l0x">github.com/pololu/vl53l0x</a> | Pololu                                      | FREE             | FREE             |
| 6  | JSON (built-in)        | Library       | Serialising and saving the 2D grid map to storage at each checkpoint                    | <a href="https://python.org">python.org</a>                               | Python Software Fou                         | FREE             | FREE             |
| 7  | Python 3               | Software Tool | Main programming language for Jetson Orin Nano high-level software                      | <a href="https://python.org">python.org</a>                               | Python Software Foundation                  | FREE             | FREE             |
| 8  | JetPack SDK            | Software Tool | NVIDIA SDK for Jetson Orin Nano including CUDA, cuDNN, and TensorRT                     | <a href="https://developer.nvidia.com">developer.nvidia.com</a>           | NVIDIA                                      | FREE             | FREE             |
| 9  | Arduino Framework      | Library       | Core firmware structure, GPIO handling, interrupt service routines for encoder counting | <a href="https://arduino.cc">arduino.cc</a>                               | Arduino.cc                                  | FREE             | FREE             |
| 10 | ESP32Servo             | Library       | PWM servo control for MG996R rescue kit deployment mechanism                            | <a href="https://github.com/madhephaestus">github.com/madhephaestus</a>   | Kevin Harrington                            | FREE             | FREE             |
| 11 | HardwareSerial         | Library       | UART communication with mBot2 colour sensor using mBlock packet protocol                | <a href="https://arduino.cc">arduino.cc</a>                               | Espressif / Arduino                         | FREE             | FREE             |
| 12 | Visual Studio Code     | Software Tool | Primary IDE for Python development on Jetson and firmware development                   | <a href="https://code.visualstudio.com">code.visualstudio.com</a>         | Microsoft                                   | FREE             | FREE             |
| 13 | Arduino IDE            | Software Tool | IDE for ESP32-S3 firmware development and upload                                        | <a href="https://arduino.cc">arduino.cc</a>                               | <a href="https://arduino.cc">Arduino.cc</a> | FREE             | FREE             |
| 14 | Fusion 360             | Software Tool | 3D CAD design for chassis, brackets, camera mounts, and rescue kit mechanism            | <a href="https://autodesk.com/fusion360">autodesk.com/fusion360</a>       | Autodesk                                    | FREE (Education) | FREE (Education) |
| 15 | Git / GitHub           | Software Tool | Version control for managing source code and documentation                              | <a href="https://github.com">github.com</a>                               | GitHub / Microsoft                          | FREE             | FREE             |